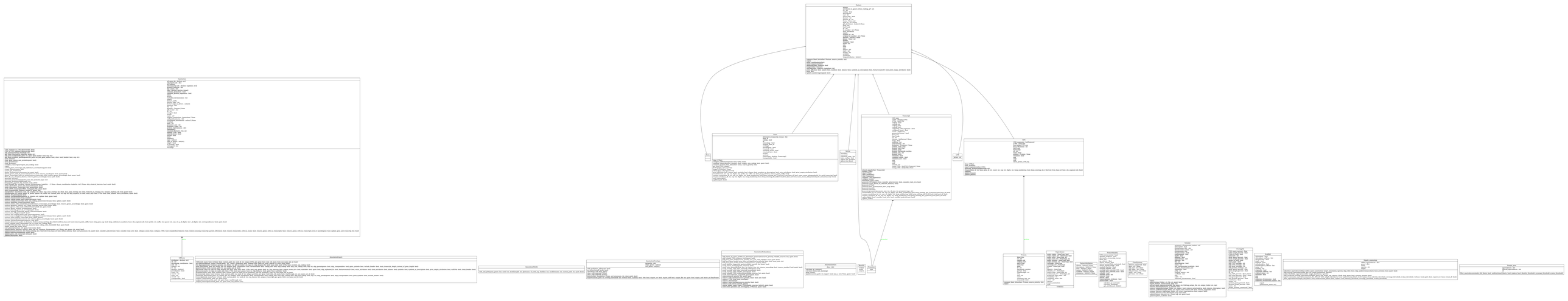

Feature
aliases
attributes_to_ignore_when_reading_gff : set
ch : str
coding : bool
descriptors
end : int
extra_copy : bool
feature : str
feature_id : str
frame : None | int
gene_id : str | None
gtf_attributes : list[str] | None
hard_seq
hard_seqs
id : str
id_number : int | None
misc_attributes
names
original_id : str
original_id_number : int | None
parents : list[str] | None
phase : None | int
quality
renamed : bool
score : str
seq
seqs
size
source : str
start : int
strand : str
symbols
synonyms
temp_attributes : list[str]
compare_blast_hits(other: Feature, source_priority: list)
copy()
equal_coordinates(other)
equal_sequence(other)
identical(other: Feature): bool
longer(other: Feature)
overlap(other: Feature): tuple[bool, int]
print_gff(clean: bool, names: bool, symbols: bool, aliases: bool, symbols_as_description: bool, featurecountsID: bool, print_empty_attributes: bool)
print_gtf()
update_numbering(original: bool)
Annotation
all_gene_ids : dict[str, str]
all_protein_ids : dict
all_suffixes
all_transcript_ids : dict[str, tuple[str, str]]
atypical_features : list
bar_colors : list
chrs : dict[str, dict[str, Gene]]
contains_promoters : bool
contains_protein_sequences : bool
entry
errors
excluded_chromosomes : list
export
feature_suffix
feature_tags : set
feature_tags_to_detect : set[str]
features : dict
file : str
genome : Genome | None
gff_header : list
id : str
merged : bool
motifs
name : str
original_annotation : Annotation | None
orphaned_features : list
overlapped_annotations : set[str] | None
overlaps
path : str
promoter_size : int
promoter_types : str
protein_equivalences : dict
redundancy
renamed_features : list, set
shared_UTRs : bool
shared_exons : bool
sorted : bool
stats
summary
tags : set[str]
tags_to_detect : set[str]
target : bool
to_overlap : bool
unmapped : list
warnings
CDS_segment_to_CDS_ids(override: bool)
CDS_to_CDS_segment_ids(override: bool)
add_aliases(overlap_threshold: int)
add_blast_hits(source, blastfile, mode: str)
add_gene_symbols(file_path: str, clear: bool, header: bool, sep: str)
add_gene_symbols_pseudogenes(file_path: str, just_gene_names: bool, clear: bool, header: bool, sep: str)
clear_aliases()
clear_gene_names_and_symbols(quiet: bool)
clear_promoters()
clear_proteins()
combine_transcripts(respect_non_coding: bool)
copy()
correct_gene_transcript_and_subfeature_coordinates(quiet: bool)
create_featurecounts_ids()
create_gtf_attributes()
define_synteny(sort_processes: int, quiet: bool)
detect_genes_with_no_transcripts(remove: bool, remove_pseudogene: bool, quiet: bool)
detect_transcripts_with_no_exons(remove_transcripts: bool, remove_genes_accordingly: bool, quiet: bool)
filter_by_rna_class(rna_classes, remove_genes_accordingly: bool, quiet: bool)
generate_introns()
generate_promoters(promoter_size: int, promoter_type: str)
generate_proteins(readthrough: str)
homogenise_parents_for_shared_exons_utrs()
load_data(gff_file, encoding, chosen_chromosomes: tuple[str, ...] | None, chosen_coordinates: tuple[int, int] | None, skip_atypical_features: bool, quiet: bool)
make_alternative_genes_into_transcripts(quiet: bool)
make_alternative_transcripts_into_genes(quiet: bool)
mark_rRNA_transcripts(rRNA_transcripts_file, clean: bool)
mark_transposable_element_genes(TE_genes_file)
merge(other: Annotation, max_cds_overlap: int | float, max_exon_overlap: int | float, max_gene_overlap: int | float, features_to_rename: list, rename_clashing_ids: bool, quiet: bool)
release(name, id, source_name, id_prefix, spacer: int, suffix: str, custom_path: str, tag: str, skip_atypical_fts: bool, main_only: bool, UTRs: bool, clear_aliases, extra_attributes, quiet: bool)
remove_TE_genes(quiet: bool)
remove_chromosomes(features_to_remove: set, update: bool, quiet: bool)
remove_chromosomes_from_header()
remove_coding_genes_and_transcripts(quiet: bool)
remove_coding_transcripts_from_non_coding_genes(removed_any: bool, update, quiet: bool)
remove_duplicate_transcripts(quiet: bool)
remove_exons_with_unmatched_strand(remove_transcripts_accordingly: bool, remove_genes_accordingly: bool, quiet: bool)
remove_genes(to_remove: set | None, override_rescue: bool, quiet: bool)
remove_genes_with_small_CDSs(CDS_threshold: int, quiet: bool)
remove_genes_without_symbols(quiet: bool)
remove_missing_genes_in_overlaps(quiet: bool)
remove_missing_transcript_parent_references(quiet: bool)
remove_non_TE_genes(quiet: bool)
remove_non_coding_genes_and_transcripts(quiet: bool)
remove_non_coding_transcripts_from_coding_genes(removed_any: bool, update, quiet: bool)
remove_other_mRNA_transcripts_from_rRNA_genes()
remove_transcripts(to_remove: set, remove_genes_accordingly: bool, quiet: bool)
rename_chromosomes(equivalences, dap: bool)
rename_ids(custom_path: str, features: list[str], keep_existing_ids_if_derived_from_base_id: bool, remove_point_suffix: bool, strip_gene_tag: bool, keep_subfeature_numbers: bool, cds_segment_ids: bool, prefix: str, suffix: str, spacer: int, sep: str, g_id_digits: int, t_id_digits: int, correspondences: bool, quiet: bool)
return_random_gene_ids(number: int, to_avoid: list, coding: bool)
rework_CDSs(override: bool, low_memory: bool, coding_ratio_threshold: float, quiet: bool)
single_parent_for_exons_utrs()
sort_genes(processes: int, quiet: bool, noisy: bool)
subset(chosen_features: set[str], gene_cap: int, common_chromosomes: set | None, min_genes: int, quiet: bool)
update(rename_features: list, keep_existing_ids_if_derived_from_base_id: bool, define_synteny: bool, sort_processes: int, quiet: bool, consider_polycistronic: bool, consider_read_utrs: bool, collapse_exons: bool, collapse_CDSs: bool, standardise_features: bool, remove_missing_transcript_parent_references: bool, remove_transcripts_with_no_exons: bool, remove_genes_with_no_transcripts: bool, remove_genes_with_no_transcripts_even_if_pseudogene: bool, update_gene_and_transcript_list: bool)
update_features(standardise, quiet: bool)
update_gene_and_transcript_list(quiet: bool)
update_keys(quiet: bool)
entry
Exon
Gene
alternative_transcript_rescue : list
base_id
coding : bool
id
noncoding : bool
original_base_id
overlaps
pseudogene : bool
renamed : bool
renamed_exons : bool
renamed_utrs : bool
strand
synteny
transcripts : dict[str, Transcript]
transposable : bool
clear_UTRs()
collapse_subfeatures(exons: bool, CDSs: bool)
combine_transcripts(low_memory: bool, respect_non_coding: bool, quiet: bool)
compare_protein_blast_hits(other: Gene, source_priority: list)
get_main_CDS_range()
homogenise_exon_scores()
longer_CDS(other: Gene)
obtain_base_id(original: bool)
print_gff(clean: bool, names: bool, symbols: bool, aliases: bool, symbols_as_description: bool, extra_attributes: bool, print_empty_attributes: bool)
rename(count: int, sep: str, digits: int, prefix: str, suffix: str, base_id_as_id: bool, remove_point_suffix: bool)
rename_exons(base_id: str, sep: str, digits: int, keep_numbering: bool, keep_existing_ids_if_derived_from_base_id: bool, name_exons_independently_for_each_transcript: bool)
rename_utrs(base_id: str, sep: str, digits: int, keep_numbering: bool, keep_existing_ids_if_derived_from_base_id: bool, name_exons_independently_for_each_transcript: bool)
sort_transcripts()
update(quiet: bool)
Intron
boundary
canonical
canonical_seqs : list
intra_coding : bool
splice_site_acceptor
splice_site_donor
Transcript
CDS_size
CDSs : dict[str, CDS]
UTRs : list[UTR]
coding : bool
coding_end
coding_ratio
coding_start
collapsed_CDS_segments : bool
collapsed_exons : bool
exons : list[Exon]
generated_exons : bool
hard_seq
hard_seqs
id : str
introns : list[Intron] | None
main : bool
miRNAs : list
polycistronic : str
promoter : Promoter | None
protein_early_stop
protein_end_stop
protein_gaps
protein_nucleotide_surplus
protein_seq : str
protein_start
renamed : bool
renamed_exons : bool
renamed_utrs : bool
seq
seqs
size
strand : str
temp_CDSs : list[CDS | Feature] | None
temp_UTRs : list[UTR] | None
almost_equal(other: Transcript)
assign_UTRs()
clear_UTRs()
clear_promoter()
clear_sequence()
collapse_CDS_segments()
collapse_exons()
determine_main_CDS()
generate_CDSs(quiet: bool, consider_polycistronic: bool, consider_read_utrs: bool)
generate_CDSs_based_on_ORF(low_memory: bool)
generate_UTRs()
generate_best_protein(must_have_stop: bool)
generate_exons()
generate_introns()
generate_promoter(promoter_size: int, ch_size: int, promoter_type: str)
rename(base_id: str, count: int, sep: str, digits: int, keep_numbering: bool, keep_existing_ids_if_derived_from_base_id: bool)
rename_exons(base_id: str, sep: str, digits: int, keep_numbering: bool, keep_existing_ids_if_derived_from_base_id: bool)
rename_utrs(base_id: str, sep: str, digits: int, keep_numbering: bool, keep_existing_ids_if_derived_from_base_id: bool)
update(quiet: bool, consider_read_utrs: bool, consider_polycistronic: bool)
update_UTRs()
promoter
UTR
prime : str
CDS
CDS_segments : list[Feature]
UTRs : list[UTR]
five_prime_UTR_seq
full_UTR_exons : int
hard_seq
hard_seqs
id : str
main : bool
protein : Protein | None
renamed : bool
seq
seqs
size
three_prime_UTR_seq
clear_UTRs()
clear_protein()
equal_segments(other: CDS)
generate_protein(readthrough: str)
rename(base_id: str, base_gene_id: str, count: int, sep: str, digits: int, keep_numbering: bool, keep_existing_ids_if_derived_from_base_id: bool, cds_segment_ids: bool)
update()
update_frame()
update_phase()
protein
GffEntry
attributes : dict[str, str]
ch : str
decreasing_coordinates : bool
end : int
feature : str
id : str
parents : list[str]
pseudogene : bool
score : str
source : str
start : int
strand : str
transposable : bool
AnnotationExport
CDSs(only_main: bool, verbose: bool, custom_path: str, used_id: str, unique_CDSs_per_gene: bool, only_cds_main: bool, use_name_not_id: bool)
all_features(feature_output: Literal['main', 'all', 'both'], promoters: bool, verbose: bool, path: str, most_specific_id_level, quiet: bool)
for_dapseq(genome: Genome, genome_out_folder: str, gff_out_folder: str, tag: str, skip_atypical_fts: bool, main_only: bool, UTRs: bool, exclude_non_coding: bool)
gene_list(custom_path: str, output_file: str, lengths: bool, coordinates: bool, chromosomes: bool, coding_info: bool, skip_coding: bool, skip_non_coding: bool, sep: str, skip_pseudogenes: bool, skip_transposables: bool, gene_symbols: bool, include_header: bool, main_transcript_length_instead_of_gene_length: bool)
genes(verbose: bool, custom_path: str, use_name_not_id: bool)
gff(custom_path: str, tag: str, skip_atypical_fts: bool, main_only: bool, UTRs: bool, just_genes: bool, no_1bp_features: bool, repeat_exons_utrs: bool, subfolder: bool, quiet: bool, skip_orphaned_fts: bool, featurecountsID: bool, extra_attributes: bool, clean_attributes: bool, aliases: bool, symbols: bool, symbols_as_description: bool, print_empty_attributes: bool, miRNAs: bool, clean_header: bool)
gtf(custom_path: str, tag: str, main_only: bool, UTRs: bool, just_genes: bool, no_1bp_features: bool, quiet: bool, subfolder: bool)
promoters(only_main: bool, verbose: bool, custom_path: str, used_id: str, promoter_type: str, use_name_not_id: bool)
proteins(only_main: bool, verbose: bool, custom_path: str, used_id: str, unique_proteins_per_gene: bool, only_cds_main: bool, readthrough: str, use_name_not_id: bool)
transcript_list(custom_path: str, output_file: str, lengths: bool, coordinates: bool, chromosomes: bool, coding_info: bool, skip_coding: bool, skip_non_coding: bool, sep: str, skip_pseudogenes: bool, skip_transposables: bool, gene_symbols: bool, include_header: bool)
transcripts(only_main: bool, verbose: bool, custom_path: str, used_id: str, rna_classes: list, unique_transcripts_per_gene: bool, use_name_not_id: bool)
unique_CDSs(custom_path: str, quiet: bool)
unique_proteins(custom_path: str, quiet: bool, readthrough: str)
unique_transcripts(custom_path: str, quiet: bool, rna_classes: list)
AnnotationMotifs
find_and_plot(query_genes: list, motif: str, motif_length: int, glistname, tf_motif_tag, backlist: list, backlistname: str, custom_path: str, quiet: bool)
AnnotationOverlaps
networks : dict
other_genes : set[str]
self_genes : set[str]
add_qualitative_info(quiet: bool)
as_networks(self_mode: bool)
clear(keep_self, keep_other)
clear_with_selected_CDSs()
clear_with_selected_exons()
detect(other: Annotation | None, sort_processes: int, clear, quiet: bool)
export(custom_path: str, overlap_threshold: int, verbose: bool, synteny: bool, NAs: bool, export_csv: bool, export_self: bool, output_file: str, quiet: bool, copies_info: bool): pd.DataFrame
AnnotationRedundancy
add_better_ab_initio_models_as_alternative_transcripts(source_priority, reliable_sources: list, quiet: bool)
add_reliable_CDS_evidence_score(quiet: bool)
find_best_gene_model(source_priority: list, just_with_reliables: bool, quiet: bool)
find_best_gene_model_exon_num_overlaps(source_priority, blast: bool, exon_num: int)
find_best_gene_model_nested_overlaps(source_priority, blast)
mark_abinitio_supported_genes(reliable_sources: list, quiet: bool)
mark_intron_nesting(ignore_removed: bool)
mark_noisy_genes(protein_size: int, intron_size: int, remove_noncoding: bool, remove_masked: bool, quiet: bool)
mark_overlap_with_other_selected_CDSs(quiet: bool)
mark_overlap_with_other_selected_exons(quiet: bool)
mark_overlap_with_reliable_genes(quiet: bool)
mark_reliable_CDS_evidences(unreliable_sources: list, quiet: bool)
mark_transcriptomic_supported_genes(quiet: bool)
remove(source_priority: list, quiet: bool)
remove_CDS_overlaps(source_priority, blast: bool, anti: bool)
remove_UTRs_from_exon_overlaps()
remove_alternative()
remove_exon_overlaps(source_priority, blast: bool)
remove_fully_intron_nested_genes()
rescue_longer_same_frame_CDSs(reliable_sources: list[str], quiet: bool)
select_best_possible_non_overlapping_UTR(exon, quiet: bool)
AnnotationStats
data : dict
calculate_gc_content()
calculate_transcript_masking()
gene_count()
update(custom_path: str, export: bool, max_x: int | None, quiet: bool)
BlastHit
evalue
score
source
Promoter
type
Protein
blast_hits
ch : str
coding_end
coding_start
early_stop
end_stop
gaps
id : str
nucleotide_surplus
partial : bool
readthrough : str
seq
size : int
start
summary_tag : str
truncated : bool
compare_blast_hits(other: Protein, source_priority: list)
copy()
Equivalence
aegis_copies : NoneType, bool
aegis_copies_rev : NoneType, bool
aegis_score : NoneType, int
aegis_score_rev : NoneType, int
aegis_synteny : NoneType, bool
aegis_synteny_rev : NoneType, bool
coverage : NoneType
coverage_rev : NoneType
evalue : NoneType, float
id
identity : NoneType
identity_rev : NoneType
match_num : NoneType, int
match_num_rev : NoneType, int
preferred_type_order : list
reliability : str
reliability_order : list
score : str
species
target_annotation
type
verbose()
FeatureAttributes
aliases : list[str] | None
descriptors : list[str] | None
misc : list[str] | None
names : list[str] | None
symbols : list[str] | None
synonyms : list[str] | None
FeatureQuality
UTR_intron_nested : bool
abinitio_evidence : bool
alternative_transcript_rescue
blast_hits
gc_content : float
intron_nested : bool
intron_nested_fully_contained : bool
intron_nested_single : bool
masked_fraction : float
overlap_reliable : bool
overlap_with_selected_CDS : bool
overlap_with_selected_exon : bool
reliable : bool
reliable_score : int
remove : bool
rescue : bool
transcriptomic_evidence : bool
unrescuable : bool
calculate_gc_content()
calculate_masking()
get_attributes(): list[str]
GeneSynteny
liftover_conserved : bool
next : str | None
old_next : str | None
old_order : int | None
old_previous : str | None
order : int | None
previous : str | None
Genome
accessory_chromosome_names : set
chloroplast : bool
chromosome_dict : dict
chromosome_names : set
chromosome_size
chromosomes : bool
confrenamed : bool
dapfit : bool
dapmod : bool
equivalences : dict
features : list
file : str
mitochondria : bool
name : str
non_chromosomal_scaffolds : bool
nuclear_chromosome_size
organelles : bool
path : str
scaffold_names : set
scaffolds : dict
size
suffix : str
unknown_chromosome : bool
copy()
export(output_folder: str, file: str, quiet: bool)
export_feature_sizes(custom_path: str)
extract_peak_sequences(output_file_name: str, DAPseq_output_file: str, output_folder: str, top)
remove_features(features_to_remove: set)
remove_organelles(output_folder: str, export: bool, remove_mitochondria: bool, remove_chloroplast: bool)
remove_scaffolds(output_folder: str, export: bool, remove_00: bool, remove_organelles: bool)
rename_features_dap(output_folder: str, return_equivalences: bool, export: bool)
rename_features_from_dic(rename_map: dict): dict
subset(chosen_features: set | None, cap: int, quiet: bool)
update(update_scaffolds: bool)
OverlapHit
CDS_query_percent : float
CDS_target_percent : float
CDSs_in_both : bool
antiscore : int
exon_query_percent : float
exon_target_percent : float
exons_in_both : bool
extra_copy : bool
full_CDS_overlaps : int
full_UTR_overlaps : int
full_exon_overlaps : int
full_protein_overlaps : int
gene_query_percent : float
gene_target_percent : float
id : str
min_CDS_percent : float | None
min_exon_percent : float | None
min_gene_percent : float
min_protein_percent : float
orientation : bool
origin : str
protein_query_percent : float
protein_target_percent : float
score : int
target_synteny_conserved : bool
Scaffold
chloroplast : bool
chloroplast_suffixes : list
chromosome : bool
dapfit : bool
dapmod : bool
mitochondria : bool
mitochondria_suffixes : list
name : str
number : str
organelle : bool
organelle_suffixes : list
original_name : str
renamed : bool
seq
size
unknown_chromosome : bool
unknown_chromosome_names : list
copy()
update(new_name: str)
Simple_annotation
added_equivalences : dict
genes : dict
name
species : str
target
add_blast_equivalences(blast_folder, query_annotation, target_annotation, species, skip_rbhs: bool, skip_unidirectional_blasts: bool, proteins: bool, quiet: bool)
add_mcscan_equivalences(file, key_col, target_annotation, species)
add_orthofinder_equivalences(file, target_annotation, species)
add_reciprocal_overlap_equivalences(folder, query_tag, target_tag, species, liftoff: bool, quiet: bool, synteny_present: bool)
export_summary_equivalences(output_file, filtered: bool, simple_rbh_blasts: bool, unidirectional_blasts: bool, replace: bool, identity_threshold, coverage_threshold, evalue_threshold, verbose: bool, quiet: bool, export_csv: bool, return_df: bool)
filter_equivalences(simple_rbh_blasts: bool, unidirectional_blasts: bool, replace: bool, identity_threshold, coverage_threshold, evalue_threshold)
Simple_gene
equivalences : list
filtered_equivalences : list
id
filter_equivalences(simple_rbh_blasts: bool, unidirectional_blasts: bool, replace: bool, identity_threshold, coverage_threshold, evalue_threshold)